Image Source: European Space Agency

China's Growing Space Debris Problem

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A Rising Threat In LEO

- China is abandoning used launcher rocket stages in low Earth orbit (LEO) at an unprecedented and increasing pace
- From January 2021 to January 2025, China abandoned 51 spent rocket bodies above 650 km in altitude
  - Accounts for a massive 86% of the global total in that timeframe
- The mass of abandoned rocket bodies from China has more than tripled in recent years
  - Significant rise from 98,000 kg to 305,000 kg
  - This represents 98% of the global increase in abandoned mass
- By contrast, in the same period.....
  - The United States left only four rocket bodies
  - Russia left only one rocket body

Image Source: Deep in Space – The Science Board

LEO Explosions: A Unique Issue

Image Source: Getty Images

- The derelicts are often second-stage or kick rockets used to push payloads into high orbits before they begin their operations
  - Left behind with residual fuel that can cause explosions
- Three Chinese rocket bodies have already exploded in the last four years
  - Two CZ-6A rockets and one Zhuque-2
  - Residual debris will linger for decades to centuries
- Chinese rockets are generally larger than those of other nations
  - Significantly more dangerous fragmentation debris when they explode or collide in orbit
- These payloads primarily include satellites for China's expanding Guowang and Qianfan constellations
  - Designed as domestic alternatives to the Starlink broadband network
  - Only 200 of a planned 15,000 satellites (!) launched so far
  - So…..many more of these to come

Some Context and Examples

- China is not the worst offender in terms of number of objects or total mass
- However, net change over the last five years is more than 30x the rest of the world
- Many of these rocket bodies are at 800 km to 850 km altitude
  - US Space Defense Agency's Proliferated Warfighter Space Architecture (PWSA) will operate at ~1000 km
- Long March 6A Rocket
  - August 2024 launch
  - Explosion resulted in over 1000 pieces of trackable debris at ~810 km altitude
- Zhuque 2E Rocket
  - June 2026 breakup
  - 100 pieces of debris
  - Direct threat to ISS and Starlink constellation

R/Bs by Country	Above 650 km in LEO					
	01Jan21		01Jan26		Net Change	
	#	kg	#	kg	#	kg
China	41	98k	96	305k	+51	+207k
Russia	437	786k	438	787k	+1	+~1k
US	167	67k	171	67k	+4	+<1k
ROW	38	37k	41	41k	+3	+4k
Total	683	988k	746	1,200k	+59	+212k

Table Source: BreakingDefense.com

Long March 6A Rocket Debris
Source: Slingshot Aerospace

International Law and Potential Solutions

Image Source: Center for Ethics and Rule of Law

- International best-practice guidelines mandate a number of mitigation measures to reduce the risks of on-orbit explosions
  - Signed by more than 60 nations including China
  - Verification of compliance is difficult
- Potential mitigator – Left Over Fuel:
  - Use left-over fuel to maneuver spent rocket stages to low altitudes
  - Stages they fall back to Earth within 25 years naturally, or….
  - Preferably, in a controlled manner
- Potential mitigator – Orbit Injection:
  - Jettison spent launcher stages in a lower orbit to ensure 25-year deorbit
  - Rely on electric or other propulsion more slowly reach operational altitude
- PRC claims to already follow international best practices (June 11, 2026 Statement to UN Committee on Peacful Uses of Outer Space)

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